
Model-Based Shape Optimization of Deformable Linear Objects via Robotic Manipulation

Research Practice – Proposed Project Structure

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This document outlines the proposed scope and structure of the Research Practice project, to establish a common basis for discussion and supervision.

Project Objective

- Study model-based methods for optimizing the shape of deformable linear objects under robotic manipulation.
- Define the optimization problem in terms of target shape, boundary actuation, robot end-effector pose, computation time, and final shape accuracy.
- Analyze the trade-off between physically accurate rod models and faster simplified or reduced-order modeling approaches.

Baseline Rod Modeling and Shape Optimization

- Use Cosserat rod modeling as the main theoretical baseline for DLO deformation.
- Formulate the shape optimization objective with elastic energy and target-shape error terms.
- Study numerical convergence, initialization sensitivity, and the possible need for continuation strategies.

Integration of the MATLAB Toolbox

- Investigate how the provided MATLAB toolbox for efficient simulation of rigid-flexible multibody systems can be included in the project.
- Focus on its relative-kinematic variational integrator formulation and its ability to reduce the model by excluding unconstrained degrees of freedom.
- Evaluate whether this modeling framework can speed up the shape optimization pipeline compared with the baseline rod optimization model.

Simplified Model for Initialization

- Study whether a simplified DLO model can provide a fast approximate shape or end-effector pose.
- Use the simplified result as a warm start for a higher-fidelity optimization model.
- Assess whether this reduces computation time while preserving the final shape accuracy.

Robotic Manipulation Interface

- Relate the optimized DLO boundary conditions or tip pose to a robot end-effector command.
- Keep the project focused on model-based shape optimization, while maintaining a clear connection to robotic manipulation.

Evaluation Criteria

The different modeling and optimization approaches will be compared according to:

- computation time;
- convergence behavior;
- final shape error with respect to the target shape;
- robustness with respect to initialization and target complexity;
- suitability for integration in a robotic manipulation pipeline.

References

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